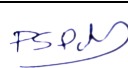
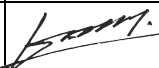
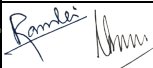


Reliance Project Management System

Standard Specification of Duct & Pipe Insulation for HVAC RPMS-ENGG-SPC-HV-019

					
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Contents

1.	Purpose.....	4
2.	References.....	4
3.	Conflicts and Deviations.....	4
4.	Abbreviations & Definitions	5
5.	Technical and Design Requirements	6
6.	Inspection and Testing	11
7.	Vendor Data	11
8.	Packing, Preservation, Shipment and Storage	11

1. Purpose

This specification defines the minimum technical requirements for design, materials, manufacture, inspection, testing, packaging, shipment and documentation of Duct & Pipe Insulation, for Supplier and/or Contractor to comply with, to achieve Owner's objective of standardization.

2. References

The latest edition of the following codes and standards shall be complied:

BS 476	Fire Tests on Building Materials & Structures
BS 2972	Inorganic Thermal Insulating Materials
BS 5970	Code of Practice for Thermal Insulation of Pipework and Equipment
BS 1470	Specification for wrought aluminum and aluminum alloys for general engineering purposes: plate, sheet and strip
BS 5422	Method for specifying thermal insulating materials for pipes, tanks, vessels, ductwork and equipment operating within the temperature range -40°C to +700°C
ASTM B209	Specification for Aluminum and Aluminum Alloy Sheet and Plate.
ASTM C 1071	Standard specification for fibrous glass duct lining insulation(Thermal and sound absorbing material)
ASTM E 84	Test method for surface burning characteristics for building materials.
ASTM C 411	Test method for Hot surface performance of high temperature thermal insulation.
ASTM C 916	Standard specifications for Adhesives for duct thermal insulation.
ASTM C 1290	Standard specifications for flexible fibrous glass blanket insulation.
IS 11239	Standard specification for Under deck insulation.

3. Conflicts and Deviations

In the event of conflict or deviation from requirements stipulated in this specification and other applicable documents, the following order of precedence shall govern:

Noted

1. Data Sheets and Drawing

2. The Material Requisition
3. This Standard Specification
4. Codes and Standards Specifically Referred
5. Other National and International Standards as Referenced

Latest revisions of codes and other statutory requirements as applicable at the time of enquiry, shall be applicable. In the event of a conflict between the requirement of the referenced codes, standards, specifications, regulations and procedures, Supplier shall submit the recommended solution to Buyer for review and permission to proceed prior to implementation.

Notes:

- “Data Sheets and Drawings” represent those attached to the Material Requisition.
- “Material Requisition” represents Material Requisition for Quotation, superseded by Material Requisition for Purchase on execution of a Purchase Order.
- Any local statutory and regulatory requirement shall mandatorily be applicable and addressed during engineering stage.

4. Abbreviations & Definitions

HSE	Health Safety and Environment
PIR	Polyisocyanurate
SWG	Standard Wire Gauge
RCC	Reinforced Cement Concrete
NB	Nominal Bore.

Parties responsible for complying with an Engineering Standard

Supplier	Party which manufactures or supplies equipment and services to perform the function as specified by Buyer
Buyer	Owner (Reliance) or Owner-appointed DEC or EPC who specifies equipment and services for purchase from Supplier

Parties referred within body of Engineering Standard

Owner	Reliance Industries Limited or specified Reliance Group Company
DEC	Detailed Engineering Contractor
EPC	Engineering, Procurement and Construction Contractor
Contractor	DEC or EPC

Noted

5. Technical and Design Requirements

5.1. Insulation Materials

Products shall not contain asbestos, lead, mercury, or mercury compounds.

Products that come in contact with stainless steel shall have a leachable chloride content of less than 50 ppm when tested according to ASTM C 871.

Insulation materials for use on austenitic stainless steel shall be qualified as acceptable according to ASTM C 795.

Foam insulation materials shall not use CFC or HCFC blowing agents in the manufacturing process.

Glass-Fiber Blanket Insulation: Glass fibers bonded with a thermosetting resin. Comply with ASTM C 1290.

Glass-Fiber Board Insulation: Glass fibers bonded with a thermosetting resin.

5.2. Aluminum Foil Tape

Product performance is based on products manufactured by Venture Tape; there are slight variations among manufacturers.

Vapor-retarder tape with cross linked acrylic adhesive.

Width: 2 inches (50 mm).

Thickness: 65Microns

Adhesion: 3.5N/CM

Elongation: 5 percent.

Tensile Strength: 34 lbf/inch (6.2 N/mm) in width.

5.3. General Installation Requirements

Install insulation materials, accessories, and finishes with smooth, straight, and even surfaces; free of voids throughout the length of ducts and fittings.

Install insulation materials, vapor barriers or retarders, jackets, and thicknesses required for each item of duct system as specified in insulation system schedules.

Install accessories compatible with insulation materials and suitable for the service. Install accessories that do not corrode, soften, or otherwise attack insulation or jacket in either wet or dry state.

Install multiple layers of insulation with longitudinal and end seams staggered.

Keep insulation materials dry during application and finishing.

Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer.

Install insulation with least number of joints practical.

Noted

Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic. Install insulation continuously through hangers and around anchor attachments. For insulation application where vapor barriers are indicated, extend insulation on anchor legs from point of attachment to supported item to point of attachment to structure. Taper and seal ends at attachment to structure with vapor-barrier mastic. Install insert materials and install insulation to tightly join the insert. Seal insulation to insulation inserts with adhesive or sealing compound recommended by insulation material manufacturer.

Apply adhesives, mastics, and sealants at manufacturer's recommended coverage rate and wet and dry film thicknesses.

Cut insulation in a manner to avoid compressing insulation more than 75 percent of its nominal thickness. Finish installation with systems at operating conditions. Repair joint separations and cracking due to thermal movement.

Repair damaged insulation facings by applying same facing material over damaged areas. Extend patches at least 4 inches (100 mm) beyond damaged areas. Adhere and seal patches similar to butt joints.

Self-Sticking-Base Insulation Hangers: Baseplate welded to projecting spindle that is capable of holding insulation, of thickness indicated, securely in position indicated when self-locking washer is in place. Base plate: Galvanized carbon-steel sheet, 0.030 inch (0.76 mm) thick by 2 inches (50 mm) square. Spindle: [Galvanized carbon-steel], fully annealed, 0.106-inch- (2.6-mm-) diameter shank, length to suit depth of insulation indicated. Adhesive-backed base with a peel-off protective cover. Insulation-Retaining Washers: Self-locking washers formed from 0.016-inch- (0.41-mm-) thick, [galvanized-steel] sheet, with beveled edge sized as required to hold insulation securely in place but not less than 1-1/2 inches (38 mm) in diameter.

See Editing Instruction No. 1 in the Evaluations for cautions about naming manufacturers and products. See Section 016000 "Product Requirements."

See Editing Instruction No. 1 in the Evaluations for cautions about naming manufacturers and products. Retain subparagraph and list of manufacturers below. See Section 016000 "Product Requirements."

In paragraph below, stainless steel is the most common wire used and is best suited for all applications.

See Editing Instruction No. 1 in the Evaluations for cautions about naming manufacturers. Retain subparagraph and list of manufacturers below. See Section 016000 "Product Requirements."

5.4. Duct Thermal Insulation

Noted

Comply

Insulation material shall be resin bonded fiber glass having thermal conductivity not greater than $0.036 \text{ W / m}^{\circ} \text{ K}$ at 10°C mean temperature and should be self-extinguishing. Temperature range shall range from -40°C to 105°C . The insulation material shall have fire performance of Class 0 as per BS 476 part 6 & 7.

All material including tapes, glue shall comply with flame/smoke spread rating of 25/50. Insulation and related application materials Installed Outdoors: Flame-spread index of 75 or less, and smoke-developed index of 150 or less.

5.4.1. When Supply Duct Is In The Return Air Plenum

The supply air duct shall be externally insulated with 25 mm thick fiber glass of density 24 kg/m^3 faced with reinforced Al foil of 7 microns

Comply

The duct surface shall be cleaned and 'Foster Ductfac 81-22 adhesive/or equivalent shall be applied and allow it to dry for 5 minutes.

Stick the insulation roll (with foil-faced outside) on the duct and tie it up with 12mm wide PVC strap. Seal all joints with 50 MM wide self-adhesive aluminum tape.

5.4.2. When The Supply / Return Air Ducts Are In Attic:

The supply air and the return air duct shall be externally / thermally insulated with fiber glass – 50 mm and 25 mm thick respectively, density 24 kg/m^3 with factory reinforced Aluminum foil, wrapped with 12 mm wide PVC strap.

Comply

The duct surface shall be cleaned and 'Foster Ductfac 81-22 adhesive shall be applied and allow it to dry for 5 minutes.

Stick the insulation slab (with foil-faced outside) on the duct and tie it up with 12mm wide PVC strap. Seal all joints with 50 MM wide self-adhesive aluminum tape.

5.4.3. When The Supply / Return Air Duct Are Exposed To The Ambient:

The supply / Return air duct shall be insulated with fiber glass- 50 mm thick, density 48 kg/m^3 with factory reinforced Aluminum foil.

Prior to insulation the duct surface shall be cleaned and 'Foster Ductfac 81-22 adhesive shall be applied and allow it to dry for 5 minutes.

Stick the insulation slab (with foil-faced outside) on the duct and tie it up with 12mm wide PVC strap. Seal all joints with 50 MM wide self-adhesive aluminum tape.

The insulated duct shall be finally cladded with Aluminum sheet of 26 SWG.

5.4.4. Duct Acoustic Insulation:

Supply air duct shall be internally insulated up to 3m from the equipment with fiber glass rigid board- 15 mm thick, density 48 kg/m^3 covered with black tissue paper.

Adhesive of suitable grade shall be applied on the clean surface of the duct.

Noted

Burbed stick pins shall be glued firmly to the duct surface at every 600 mm distance.

The rigid board insulation shall be firmly placed over the pins.

Aluminum Perforated sheet of 42 SWG shall be placed over the insulation. Finally entire insulation and the Aluminum sheet shall be held in the position with the retainer Wesyer.

5.4.5. Under deck Thermal Insulation:

The under deck insulation shall be provided where roof slab of buildings are exposed to higher temperature. The proposed under deck insulation shall be CFC free, rigid, cellular foamed.

Other specifications as follows:

a) Material: Polyisocyanurate (PIR) Foams with craft paper (117 gms/sq.m; 0.19 mm thickness) on both side.

b) Type: Slab form (1 mt x 0.5 mt x 25 mm thick)

c) Density: 38 +/- 2 kg/m³

d) Compressive strength: 172 kN/Sq.m

f) Water Absorption by volume : < = 2%

g) Thermal conductivity: 0.020 W/m.deg.K at 10 deg C

Method of Application:

Under deck insulation shall be applied as follows:

A. Clean the surface to be insulated and make it free from dust and loose particles.

B. Seal the intra slab joints with PVC tapes.

C. Insulation shall be screwed to RCC slab with nylon rawl plugs using screws of the length more than the thickness of the insulation along with appropriate washers and securely tied with GI binding wire. All hardware shall be of GI material only.

5.5. Pipe Insulation

Insulation material shall be pre-formed Polyisocyanurate profiles (pipe sections) with density of 42 ± 2 kg/m³ and thermal conductivity of 0.020 W/m K° at 10 °C.

5.6. Method of Application

5.6.1. Above Ground Chilled Water Piping:

All Hydrostatic tests on piping shall be carried out before insulating materials is applied. The insulation contractor shall only insulate these sections of the pipe that has been specifically released for such work by the engineer-in charge.

The surfaces to be insulated shall be cleaned by brush & cloths to remove dust, dirt, grease etc.

Comply

Noted

A coat of bitumen primer shall be applied followed with a coat of bitumen (85/25 grade) or cold adhesive compound like CPRX or MAS-83 used as adhesive compound.

The surface shall be insulated with 30 mm / 40 mm thick Aluminum foil faced polyisocyanurate foam pipe section of density specified as above and all the joints shall be sealed with a cold adhesive.

PIR Pipe sections shall be tied up using PVC straps (box strapping) at about 500 to 600 mm distance about 2 straps every 1000 mm long pipe section.

A layer of 120 g/m³ polythene sheet, vapour barrier, shall be wrapped over the PIR pipe section.

Post insulation, 26 SWG Aluminum cladding with proper grooves and 30 mm overlaps shall cover the pipes.

Applicable Insulation thickness:

30 mm for pipe size up to 150 mm Dia. NB mm for pipe size above 150mmNB and above.

5.6.2. Under Ground Chilled Water Piping (Buried / Trench):

All Hydrostatic tests on piping shall be carried out before insulating materials is applied. The insulation contractor shall only insulate these sections of the pipe that has been specifically released for such work by the engineer-in charge.

The surfaces to be insulated shall be cleaned by brush & cloths to remove dust, dirt, grease etc.

A coat of bitumen primer shall be applied followed with a coat of bitumen (85/25 grade) or cold adhesive compound like CPRX or MAS-83 used as adhesive compound on the pipe surface & inner surface of the PIR section.

The surface shall be insulated with 30 mm / 40 mm thick Aluminum foil faced polyisocyanurate pipe section of density specified as above and all the joints shall be sealed by applying a coat of bitumen on the surface of pipe as well as PIR sections.

Ensure sealant is applied to the circumferential and longitudinal edges of the PIR pipe sections to ensure proper sealing.

PIR Pipe sections shall be tied up using PVC straps (box strapping) at about 500 to 600 mm distance about 2 straps every 1000 mm long pipe section.

A coat of 1 part CLEARSEAL & 2 parts water as primer shall be applied on the outer surface of PIR insulation and allow it to dry.

CLEARSEAL shall be applied in such a way to cover 4 to 5 Sq.m / liter. per coat and shall be allowed to dry for 3 to 4 hours.

Noted

Second coat of CLEARSEAL shall be applied and while it is sticky the pipe surface shall be wrapped with 3 mm thick fiber glass cloth and shall be allowed to dry for 3 to 4 hours.

Finally third coat of CLEARSEAL shall be applied and shall be allowed to dry for 3 to 4 hours.

Civil agency shall be then allowed to refill the trench using sand.

Insulation thickness

30 mm for pipe size up to 150 mm Dia. NB

40 mm for pipe size above 150mm NB and above

6. Inspection and Testing

Inspection and testing shall be as per Inspection and Test Plan. However, requirements specified in Material Requisition and in this specification shall be complied as a minimum.

7. Vendor Data

Vendor documentation shall be as per VDDS as enclosed with the Material Requisition.

8. Packing, Preservation, Shipment and Storage

The Supplier shall provide detailed packaging and Preservation procedure for review and approval.

The unit along with all accessories/ auxiliaries after application of preservatives suitable for a period of at least one-year storage.

The Supplier shall specify preservation requirement, if storage is beyond one year, in a separate document along with the Installation & Operation Manual.

The unit shall be packed in completely enclosed, weather proof wooden boxes to provide adequate protection during transit towards mechanical damage, exposure to atmospheric contaminants. Package size shall be such so as to facilitate easy handling by single person.

Noted